

Bohan Zhang

Email: zbohan@umich.edu, Personal Page, Google Scholar

Education

University of Michigan

Ph.D. in Information Science

Aug 2022 - Present

M.S. in Computer Science and Engineering

Aug 2020 - May 2022

Research Interests: Causal Inference, Personalization, Machine Learning

The Ohio State University

Aug 2017 - May 2020

B.S. in Computer Science and Engineering, Minor in Linguistics

Xiamen University

Aug 2015 - July 2017

Major in Computer Science and Technology (Transferred out)

Publications

1. **Bohan Zhang**, Anqi Ni, Yixin Wang, Paramveer Dhillon “Weightless Fine-Tuning: Personalizing LLMs via Logit-Space Transport.” *COLM '26*. [Link].
2. **Bohan Zhang**, Jiaxuan Li, Ali Hortaçsu, Xiaoyang Ye, Victor Chernozhukov, Angelo Ni, Eddie Huang. “Agentic Economic Modeling.” *ACL '26 Industry Track*. Link.
3. Yachuan Liu, Xiaochun Wei*, Lin Shi*, Xinnuo Li*, **Bohan Zhang**, Paramveer Dhillon, Qiaozhu Mei “ExAnte: A Benchmark for Ex-Ante Inference in Large Language Models” *Main EACL '26*. [Link].
4. **Bohan Zhang***, Yachuan Liu*, Qiaozhu Mei, Paramveer Dhillon. “Exploring Group-Level Signals for Robust Many-Domain Generalization.” *PAKDD '25*. [Link].
5. **Bohan Zhang**, Yixin Wang, Paramveer Dhillon. “Causal Inference for Human-Language Model Collaboration.” *Main NAACL '24*. [Link].
6. Kenan Alkiek, **Bohan Zhang**, David Jurgens. “Classification without (Proper) Representation: Political Heterogeneity in Social Media and Its Implications for Classification and Behavioral Analysis.” *Findings of ACL '22*. [Link].
7. **Bohan Zhang**, Prafulla Kumar Choubey, Ruihong Huang. “Predicting Sentence Deletions for Text Simplification Using a Functional Discourse Structure.” *Main ACL '22*. [Link].
8. Yingzi Bu*, Ruoxi Gao*, **Bohan Zhang***, Luchen Zhang, Duxin Sun. “CoGT: Ensemble Machine Learning Method and its Application on JAK Inhibitor Discovery.” *ACS Omega*. [Link].
9. **Bohan Zhang**, Chengke Bu, Paramveer Dhillon. “Who Owns the Text? Pairing Personas with Personalization Mitigates Ownership Loss in AI-Assisted Writing.”
10. **Bohan Zhang**, Yixin Wang, Paramveer Dhillon. “Policy Learning with Text-Based Actions: A Causal Approach.” [Link]
11. Yachuan Liu*, **Bohan Zhang***, Qiaozhu Mei, Paramveer Dhillon. “Improving Group Distributional Robustness by Learning to Rank.” [Link].
12. Xiaoyang Ye, **Bohan Zhang**, Yudi Wang, Evan Peet. “Human-Algorithm Collaboration for Personalization: Evidence from Field and GenAI Experiments.” *Amazon Economist Summit' 25*

Work and Research Experience

Applied Scientist Intern, Amazon

May 2026 – Nov 2026

- Developing a reasoning-distillation and GRPO fine-tuning pipeline that trains scalable reasoning LLMs to estimate how customer demand redistributes under catalog changes.
- Exploring RCT fine-tuning that aligns model outputs with treatment effects, grounding the LLM’s counterfactual predictions in real causal signal.
- Collaborator: Peter Belhumeur

Applied Scientist Intern, Amazon

Mar 2025 – Oct 2025

- Designed LLM-based framework aligning LLM-generated synthetic responses with small-scale human data for reliable economic inference.

- Validated the method in large-scale conjoint study and regional pricing RCTs, matching full-scale results using only 10% human samples and saving \$4 million for the company.
- Collaborators: Victor Chernozhukov and Ali Hortaçsu

Graduate Research Assistant, University of Michigan

Aug 2022 – Present

Advisor: Paramveer Dhillon

- Developed causal methods for learning dynamic policies over text-based actions in human–AI collaborative writing. Collaborator: Yixin Wang.
- Developed personalization methods for AI-assisted writing that improve writing quality while preserving users’ sense of ownership.
- Developed learning-to-rank methods for robust many-domain generalization and studied data leakage in LLM-based ex-ante inference. Collaborator: Qiaozhu Mei.

Machine Learning Software Engineer, Initium.AI Inc.

May 2021 – Aug 2021

- Developed and deployed an action-item detection model using limited labeled data.

Research Assistant, Blablalab, University of Michigan

Sep 2020 – Oct 2021

Advisor: David Jurgens

- Studied how definitions of political users affect social-media classification and behavioral analysis, including political engagement, suspicious behavior, and changes in political leaning.
- Resulted in a publication at ACL 2022.

Research Assistant, NLP Lab, Texas A&M University

May 2019 – Aug 2019

Advisor: Ruihong Huang

- Studied the relationship between functional discourse structure and sentence deletion in document-level text simplification. Developed discourse-aware models for sentence-deletion prediction
- Resulted in a publication at ACL 2022.

Instructional Assistant, The Ohio State University

Jan 2019 – May 2020

- Supported CSE 2421: Introduction to Low-Level Programming and Computer Organization.

Honors and Service

ACL Student Volunteer Award

Dean’s List of Distinguished Students

Regular Reviewer for *CL, NeurIPS

All Semesters

Skills**Programming languages**

Familiar with Python, C, Matlab; Have knowledge in C#, Java, JavaScript

Programming Tools

Pytorch, Linux, OpenGL

English Proficiency

TOEFL 107 (R:27, L:27, S:26, W:27)

GRE: V 157, Q 169